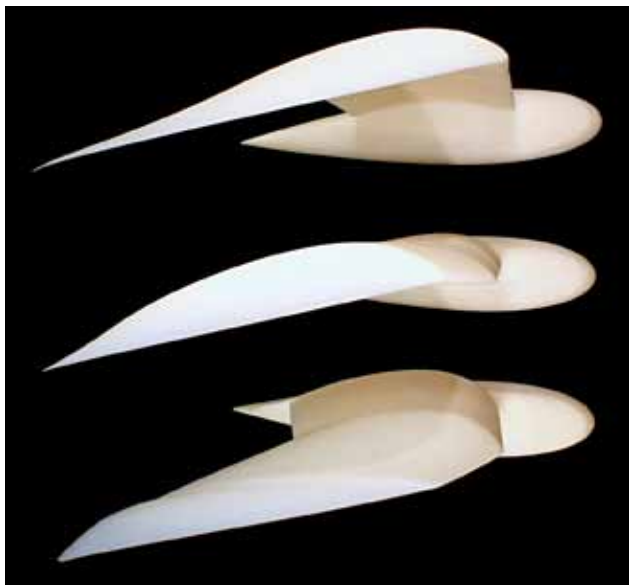




Drones being revamped to simulate bird flight

It is difficult to build unmanned aerial vehicles with the natural variability of bird wings, as birds can outsmart even the most advanced similar-sized drones. Hence,

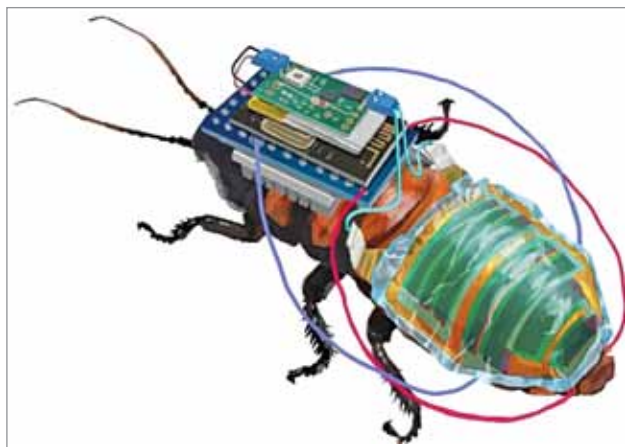


Side view of three 3D-printed gull wings that were tested in a wind tunnel to explore whether birds' complex wing shapes would be beneficial for drones (Credit: <https://techxplore.com>)

Harvey and colleagues at the University of Michigan are aiming to enhance autonomous drones' flight similar to birds' who are capable of both stable and unstable flight and use their wing movements to change the modes. The team implemented computer simulations to analyse the advantages of 3D morphing wings as they think that morphing the wings with the joint angles could enable birds to descend at steeper angles. If that's the case, 3D wing-morphing UAVs could take shorter approach paths for landing. A detailed study of bird flight can provide enhancement in drone designs for various uses.

Researchers create remote-controlled cyborg cockroaches

Investigating areas with the imminent threat has never been easier until now as researchers bring the concept of 'cyborg' into action. Researchers at RIKEN, Japan have created remote-controlled cyborg cockroaches, equipped with a tiny wireless control module that is powered by a rechargeable battery attached to a solar cell. This system was integrated into the cockroach as a backpack using an adhesion system that holds the battery and solar cell while allowing free movement to the roach. The solar cell

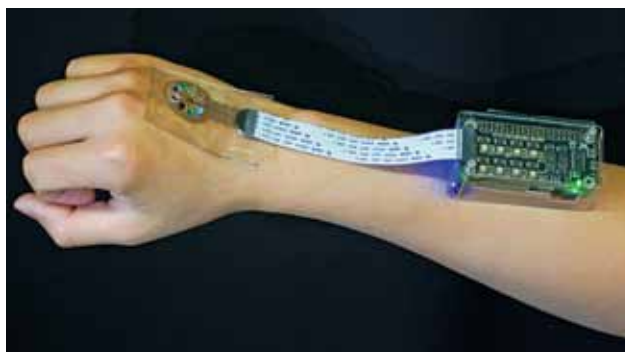


Remote-controlled cyborg cockroach (Credit: <https://scitechdaily.com>)

is 0.004mm thick and can achieve an output power of 17.2mW. Researchers plan to adapt this technology to other insects like beetles and cicadas in the future.

3D-printed light-sensitive device to help people with lupus

It is difficult to predict when a patient's symptoms will worsen after undergoing lupus disease treatment. The lupus disease worsens when exposed to sunlight or even indoor artificial light. A team of engineers and doctors at the University of Minnesota Twin Cities have invented a unique 3D-printed light-sensitive medical device to detect the progression of lupus and other light-sensitive diseases.



A unique 3D-printed light-sensitive medical device could help millions of people worldwide with lupus and other light-sensitive diseases by providing access to real-time data for more personalised treatment (Credit: <https://illinoisusnews.com>)

This fully 3D-printed device comes with a flexible UV-visible light detector that can be placed on the skin. The device is configured with a dedicated portable console for continuous monitoring and correlating lighting to symptoms, since 3D printing is a process that is inexpensive and could someday provide easy and quick access to this device.